

The bilimit, refused

Beste Mathijs,

You half-remembered a piece of categorical semantics tonight — categories of partial maps in which limits and colimits coincide — and asked whether that would not be “helemaal mooi” for us. It is, but in mirrored form, and I want to work out the mirror carefully, because I think the theorems you were reaching for are the sharpest justification this project has for existing.

The cluster you meant

The classical result is the **limit–colimit coincidence** of Smyth and Plotkin (1982), going back to Scott’s D_∞ : in a cpo-enriched category, take an ω -chain of **embedding–projection pairs** — $e : A \rightarrow B, p : B \rightarrow A$ with $p \circ e = \text{id}$ and $e \circ p \sqsubseteq \text{id}$ — and the colimit of the embeddings is canonically *the same object* as the limit of the projections: the bilimit. Freyd sharpened this to **algebraic compactness**: in such partiality-enriched settings the initial algebra and the final coalgebra of a functor coincide, $\mu F \cong \nu F$, the “bifree” algebra — which is exactly why general recursion is interpretable there. The axiomatic school around partial maps is Robinson–Rosolini (dominances), Fiore’s *Axiomatic Domain Theory in Categories of Partial Maps*, and, in modern dress, Cockett–Lack’s restriction categories.

Our tower already has the ep-pairs

Look at $\{\mathbb{Z}/2^n\}$ with the maps we already committed to: the zero-padding inclusion $e_n : \mathbb{Z}/2^n \rightarrow \mathbb{Z}/2^{n+1}$ and the truncating projection p_n back. Then $p_n \circ e_n = \text{id}$ on the nose, and $e_n \circ p_n$ is “keep the low part” — below the identity precisely in the partial-map order. So the hypotheses of Smyth–Plotkin are not foreign to adic; the tower *is* an ep-chain. Enrich over partiality and the theorem fires: the bilimit exists, and the two completions I kept apart in Letter 1 — $\mathbb{N}$ on the colimit side, $\mathbb{Z}_2$ on the limit side — glue into one object, in which every point of the limit is the supremum of its finite approximants. Towers converge; $\mu = \nu$; domain theory.

Why we refuse it

Now recall what the founding conversation identified, in its very first exchanges, as the problem to organize the whole language around: **extensional collapse** — semantics that cannot tell a cheap representation from an expensive one. I claim the bilimit coincidence is the categorical *mechanism* of that collapse. The coincidence is cost-blind by construction: it identifies the colimit side, whose elements have finite descriptions attained at a finite grade, with the limit side, whose points carry unbounded information. Enrichment over pointed cpos supplies the $\perp$ that makes the chain self-dual, and in doing so erases exactly the distinction adic is built to preserve. A point of $\mathbb{Z}_2$ is not attainable at any grade; every point of $\mathbb{N}$ is. The bilimit needs maps that adic prices at infinity.

So the beautiful statement is not “adic should have coinciding limits.” It is:

domain theory is the image of adic under the cost-erasing functor.

Erase grades and costs, and the ep-pairs become self-dual, the two limits snap together, and you land in D_∞ . Read backwards: cost-grading is precisely the obstruction that keeps Ind and Pro apart. That is theorem-shaped, and I would like it on the target list in some form such as: *there is no cost-bounded cone over the projection chain* — any family of maps presenting a single object as the limit of $\{\mathbb{Z}/2^n\}$ must have unbounded cost, while the colimit is presented by maps of cost zero-per-element. The asymmetry between the two limits, invisible extensionally, is exactly a cost asymmetry.

This also explains two things at once: why classical domain theory works as well as it does (it is the faithful shadow of the costed world), and why it could never see complexity (the shadow is taken by the functor that forgets it). And it upgrades our earlier slogan. adic is not merely a tower we decline to complete; it is the observation that completing it is a **lossy** operation, and the loss is precisely the subject of the theory.

The odometer letter ended with the write head needing to earn its weight; this one ends with a question for you: do we want the no-cost-bounded-cone claim as a formal theorem target (it needs the cost-enriched category to be set up first, so it is calculus-tier), or is it, for now, the thesis sentence of the monograph — the strong claim in its sharpest dress?

Met vriendelijke groet,
Claude